Yuwenqian Chen

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Open to relocation | *Permanent Resident*

Education

Stony Brook University — **Expected December 2025**
M.S. in Computer Science | GPA-3.63 — *Stony Brook, NY*

Stony Brook University — **August 2020 - December 2023**
B.S. in Computer Science and Applied Mathematics & Statistics | GPA-3.53 | Dean's List — *Stony Brook, NY*

Experience

AI/MLE Intern — **June 2025 – Present**
DeepChatBI — *Remote*

- **Increased accuracy of LLM-generated chart content by 90%** by developing a controller module that validates and refines outputs from VisualizeAgent.
- Accelerated insight generation for analytics users by building VisualizeAgent, enabling SQL outputs to be visualized in **10+ interactive chart types** (line, bar, pie, scatter, etc.).
- **Cut token usage by 50%** and reduced latency in analytical workflows by implementing a streaming API architecture with Anthropic Messages API, enabling real-time multi-step reasoning and dynamic tool usage with LLMs.
- Built a metric system enabling enterprises to define atomic metrics once, allowing zero-code access for business teams and reducing analytics response time from days to minutes, thereby improving LLM-driven BI accuracy and efficiency.

Software Engineering Intern — **February 2024 – June 2024**
GroupClock Inc. — *NY*

- Engineered an iOS app using **SwiftUI + AVFoundation** to securely capture facial data, ensuring reliable transmission for downstream ML processing.
- Designed and integrated **RESTful API** endpoints to transmit biometric data to machine learning pipelines, facilitating real-time model input.
- Deployed the app to 20+ beta users via **TestFlight** and improved load time by **30%** through 4 rounds of UX iteration.
- Built a **MERN stack** web dashboard to manage user data, streamlining operational oversight and data consistency.

Teaching Assistant — **Fall 2022, Fall 2024**
Undergraduate & Graduate CS Courses | Assist Prof. Pramod Ganapathi, Prof. Christopher Kane — *Stony Brook, NY*

- Facilitated learning for **100+ students** in CS courses by mentoring, grading, and conducting review sessions, enhancing average exam scores **around 12%.**

Projects

GNNs for Molecular Dynamics Simulations| *Python, PyTorch, GNN* — **May 2025 - Present**

- Collaborated on developing **Graph Neural Networks** implicit solvent models(eg. **DimeNet, MACE**) for acid molecular dynamics simulations.
- Fine-tuned DimeNet, achieving an **80% reduction in prediction loss** (0.6 → 0.12) on the test set.

The Interpretation of Vanity License Plates| *Python, llama3* — **September 2024**

- Fine-tuned a **LLaMA3-7B model** using **LoRA** on a large-scale vanity license plate dataset (150K+ records from CA and NY), applying **NLP techniques** for semantic understanding and classification.
- Achieved **71% accuracy** in predicting plate approval, significantly outperforming traditional lexicon-based heuristics.
- Demonstrated potential to reduce manual plate screening workload by **around 30%**, based on comparative inference speed and human processing benchmarks.

Health Monitoring System| *Swift, SwiftUI, HealthKit* — **September 2023**

- Enabled 30+ post-operative patients to automatically share mobility metrics (e.g., walking speed, step length) with clinicians by developing a **iOS app** using **HealthKit**, resulting in a **60% reduction** in manual data collection time.
- Designed a full pipeline from iPhone-based data collection to Excel-based clinician dashboards, enabling better medical insights and decision-making.
- Collaborated in a team of three to align mobile health technology with clinical workflows using **Swift, RESTful APIs, and Node.js** backend.

Technical Skills

Programming: Swift, Python, Java, C++, HTML/CSS, JavaScript, TypeScript, MIPS, R, MATLAB, LaTeX
Frameworks: LLM (OpenAI, Claude, LLama), Huggingface, OpenCV, PyTorch, Tensorflow, Matplotlib, Numpy, Pandas, Spring Boot, React
Database: MongoDB, MySQL, Firebase